

Graph Theory Report Guidelines

Discrete Mathematics - Spring 2026 Project (Chapter 10 - Rosen)

1. Introduction

The introduction should briefly explain graph theory and its importance in modeling relationships between objects. Students should clearly state the purpose of the report and identify the selected project topic (Graph Models, Graph Representation, or Hamiltonian Graphs).

The implementation should include a brief overview of how graphs are used in real-world systems, such as networks, transportation, and computing.

2. Problem Definition

Students must clearly define the problem they are addressing. This section should explain the real-world motivation for the chosen topic and why graph theory is well-suited to solving it.

The implementation should describe the input problem, expected output, and why traditional methods are insufficient.

3. Theoretical Background

This section should present the required theoretical concepts from graph theory. Students should define key terms such as vertices, edges, paths, connectivity, adjacency matrix, or Hamiltonian properties depending on the selected topic.

The implementation should be supported with simple explanations and diagrams where necessary.

4. Literature / Concept Review

Students should briefly discuss related concepts from the textbook (Rosen, Chapter 10) or other academic sources. This includes explaining existing graph models, representations, or algorithms.

The implementation should show understanding of prior knowledge before applying it to the chosen problem.

5. Methodology / Approach

This section explains how the problem will be solved.

Implementation should include:

- Chosen graph model or algorithm
- Step-by-step explanation of the approach
- Reason for selecting this method

Students should clearly describe how the solution will be built logically.

6. Projects

Project I: Graph Models

Students selecting this topic should describe how graphs represent real-world systems such as social networks, transportation systems, or communication networks.

The implementation should explain how vertices represent entities and edges represent relationships, and how connectivity is checked in the system.

Project II: Graph Representation

Students selecting this topic should explain different graph representation techniques, such as adjacency matrix, adjacency list, and edge list.

The implementation should compare these methods and explain the conversion between representations, especially from the adjacency list to the matrix.

Project III: Hamilton Paths and Circuits

Students selecting this topic should define Hamiltonian graphs and explain the difference between Hamiltonian paths and circuits.

The implementation should describe how the algorithm checks whether a graph contains a Hamiltonian path or circuit.

7. Implementation / Algorithm Description

Students should describe the algorithm used in the project's coding component.

The implementation should include:

- Algorithm steps
- Logic explanation
- Input-output behavior

Code is not required here, but must be explained clearly.

8. Results

This section should present the implementation's output.

The implementation should include:

- Program output
- Graph visualization (if available)
- Interpretation of results

9. Discussion

Students should analyze the results and discuss their approach's performance.

The implementation should include:

- Strengths of the method

- Limitations
- Possible improvements

10. Applications

This section should describe real-world applications of the selected topic.

Examples include:

- Social networks
- GPS navigation systems
- Computer networks
- Biological systems
- Scheduling problems

Students should explain how graph theory is used in each case.

11. Conclusion

The conclusion should summarize the entire report.

The implementation should highlight the importance of graph theory and how the selected method efficiently solves real-world problems.

12. References

Students should list all references in IEEE or standard academic format.

Examples include:

- Kenneth H. Rosen, *Discrete Mathematics and Its Applications*
- GeeksforGeeks Graph Theory Articles
- Academic papers or lecture notes